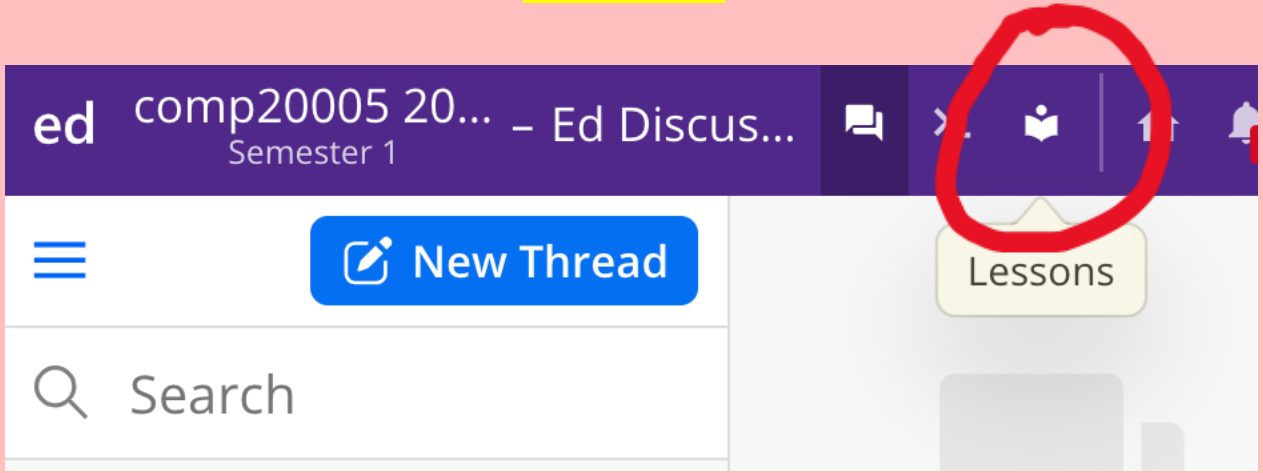
COMP20005 Workshop Week 2

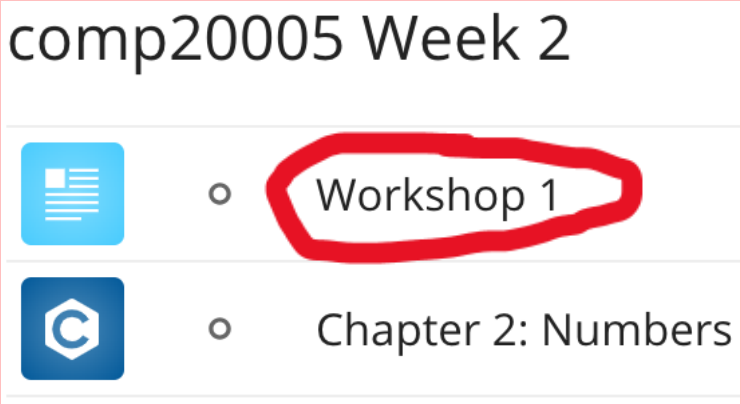
about Us and … C

While waiting:

- Talk to classmates, make friends
- Open LMS and click on "Ed Discussion" to open Ed, on Ed:

  ➢ click on Lessons then on the lesson Workshop 1





  ➢ Inspect the first slide: Welcome to the workshops!

Workshops: Learning-By-Doing

in your own time

W = 1

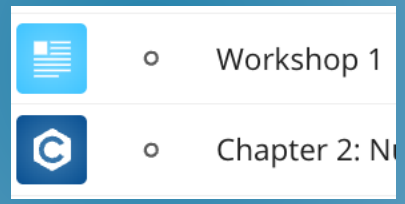
Attend & Revise lectures of week W

W = **W** + 1

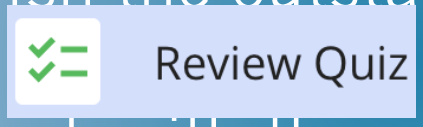
- Prepare with

LMS → Weekly Schedule → Week **W** → Workshop

in ED:



- Learn from lectures, books, **ChatGPT, Google...**

- Finish the outstanding tasks
- Do  in ED's Week W
- Check with the solutions provided

in workshop

Tutorials:

- discuss lecture materials
- be active in discussions

5-minute break

Lab time:

- be cooperative
- ask questions or say something exciting
- raise your hand to summon Anh

Lectures W1 & This Workshop

Computers & Programs

- A Computer Diagram
- Program & Programming
- C & C Compilers
- Process of solving problem in C

Inputs and Outputs

- Identifiers, variables, constant
- Types
- Operators & Precedence
- scanf & printf

This Workshop Exercises

- review variables, data types, Ex 2.01, 2.02
- Work on 1.02, 2.04, 2.08 and others from Chapters 1 and 2

Your program to solve the equation ax+b=0 (step-by-step algorithm for the Literal-Minded Computers)

```
Start
   1. do ???



Stop
```

A computer program

Problem:	**Solve equation ax + b = 0**

Program:	**Start**
	`input value for a and b;`
	`x= -b/a;`
	`output value of x;`
	End

Memo:	A typical computer program has 3 sections: 1. inputting *data* 2. computing *solution* 3. outputting *solution* *Each data need a variable, which specifies a memory container to keep the data value.*

- re-write the program in C (perhaps **using Ex 1.02**) ,
- discuss variables, data types, Ex 2.01, 2.02

Full C program: equation.c

	/* Solving equation ax + b = 0 Author: Anh Vo – avo@unimelb.edu.au ... */
Opening	#include <stdio.h> int main (int argc, char *agrv[]) {

Declaring
 	
Inputting
 	
Computing
 	
Outputting
	double a, b, x;
	
	// inputs a and b
	printf (”Enter value of a and b: ");
	scanf(”%lf %lf", &a, &b);
	
	// to do: make sure that a != 0
	// computes x as solution to ax+b= 0
	x= -b/a;
	
	// outputs result
	printf("Solution x= %lf\n", x);

Closing	return 0; }

Sample Process of Building and Running a C Program

type/edit program.c using Ed Text Editor

in Terminal, compile program.c to executable file program, using

```
[user@sahara ~]$ gcc -Wall -o program program.c
```

run the executable file program, using

```
[user@sahara ~]$ ./program
```

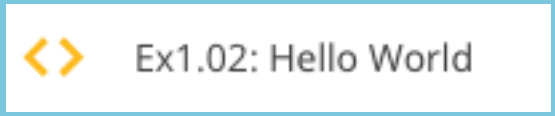
got correct outputs?

NO

YES

Note: We can change the name "program" to any reasonable name

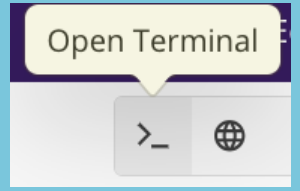
Example of using Ed: Exercise 1.02 and 2.03

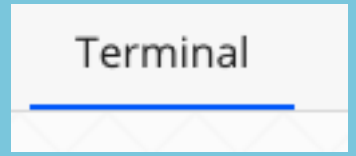
- In the lesson “Week 1 Welcome”, Click on 

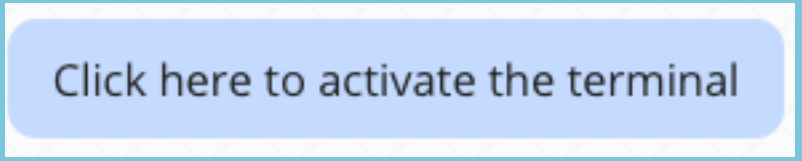
- In the window “program.c”, type your code
- Compiling and do test run using one of two methods (**Method A** or **Method B**)
- Click on

to have automatic marking of your code (and get a green tick)

Method A: compile and run the code manually

- Click on 

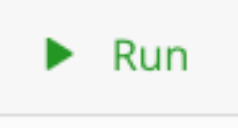
or 



- On the terminal, issue commands for compiling and running

```
[user@sahara ~]$ gcc -o program program.c
[user@sahara ~]$ ./program
```

Method B: ditto but automatically:

- Just click on 

- How it works: it uses the Makefile for automatic compiling
- Pitfall: this method doesn’t work if you have more than one .c files in the directory

[Time permits] Discuss Ex 2.01, 2.02

Variables:

- names, Ex 2.01
- data types, declaration, assignment: Ex 2.02

Lab

Discuss Ex `2.08`

Using `Ed` to complete:

Exercise `1.02` in the lesson `"Week 1: Welcome")`

Exercise `2.08, 2.04`, and others in " `Chapter 2: ...`"

- Cooperate!
- Raise hand after finishing 2.08 for Anh to come and give comments.
- Raise hand for questions or to share excitement!
- All done? Then do Review Quiz

Remember: *Programming is fun!*

Finish outstanding exercises, complete Review Quiz by this Saturday

- Stay safe, stay active, stay happy!
- Use Lectures, Books, LMS, Ed
- Wisely use Google, GenAI, . . .

Wrap-Up:

- Variables: names, data types, values
- Input with printf and scanf, output with printf
- Data types and respective formats for printf, scanf:

type	`int`	`float`	`double`	`char`	*`string`*
printf format	`%d`	`%f`	`%lf`	`%c`	`%s`
scanf format	`%d`	`%f`	`%lf`	`%c`	`%s`
scanf for **v**	`&v`	`&v`	`&v`	`&v`	`v`